

PHAROS

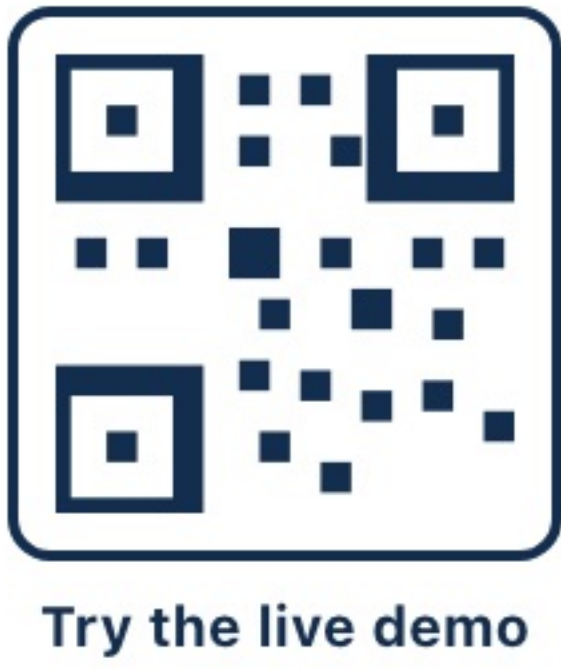
Runs in a browser

\$0 hardware

Pharos: Zero-Install Pool-Safety Detection on a Phone You Already Own

Stillness & submersion alerts, measured across real pools and published in the open

Vihaan Kommireddy, Aryan Mahalingham, Sree Dhanvin
Independent student research · Cary, North Carolina · [Science-fair & Congressional App Challenge 2026](#)



Overview

Every camera that watches a pool starts with a **\$1,269–\$1,299 hardware install**, and not one vendor publishes how often it fails. **Pharos runs on the phone in your pocket** — no install — and raises an alarm when a swimmer goes **still** too long or **slips under and does not resurface**. It is the first pool detector measured across many real pools, in the open.

Key Contributions

- 1

Zero-install detection
Runs in a browser on a phone families already own. No hardware, no mount, no calibration rig — the differentiator ten funded competitors cannot copy without cannibalizing their own product.
- 2

Measurable triggers, not “distress”
Alarms on **stillness past a threshold** and a **track that submerges and never returns** — signals a camera can measure and a human can verify, never a subjective classifier.
- 3

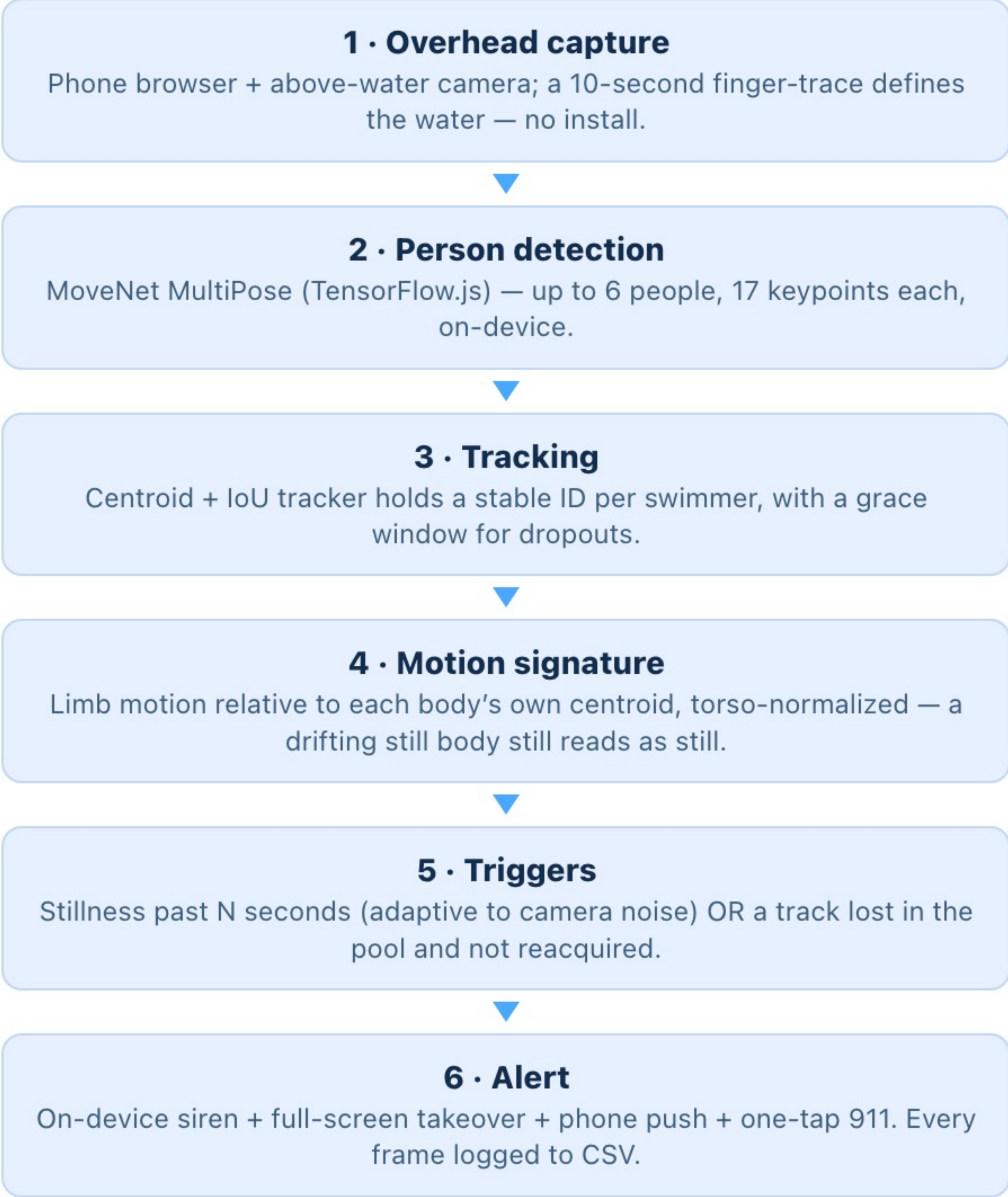
First cross-pool benchmark
An independent detection accuracy measured across multiple real pools and published — the number every vendor omits and no one else releases.
- 4

An honest negative result
We show that off-the-shelf AI is blind to people in water — then fine-tune a detector to fix it, and publish the before/after.

Problem & Motivation

Drowning is fast, silent, and the **leading cause of death for U.S. children ages 1–4** (CDC); in a 1,537-case CPSC analysis, **80% of drowned toddlers were alone**. Ten companies in five countries sell AI pool cameras, but each is a fixed **\$1,299 install tied to one pool**, and every “99%” claim is self-report with **no false-alarm rate, miss rate, or time-to-alert**. The people most exposed to drowning are the least able to buy the hardware — and no one can compare systems that publish nothing.

Method



Key Features

Runs in a browser
No app store, no install — the URL is the setup.

On-device
Video never leaves the phone; nothing uploaded.

Stillness timer
Counts seconds a swimmer stops moving.

Submersion trigger
Fires when a track goes under and never returns.

Measurement harness
Per-frame + per-alarm CSV — the published numbers.

One-tap 911
Prominent call button; never auto-dials.

Results

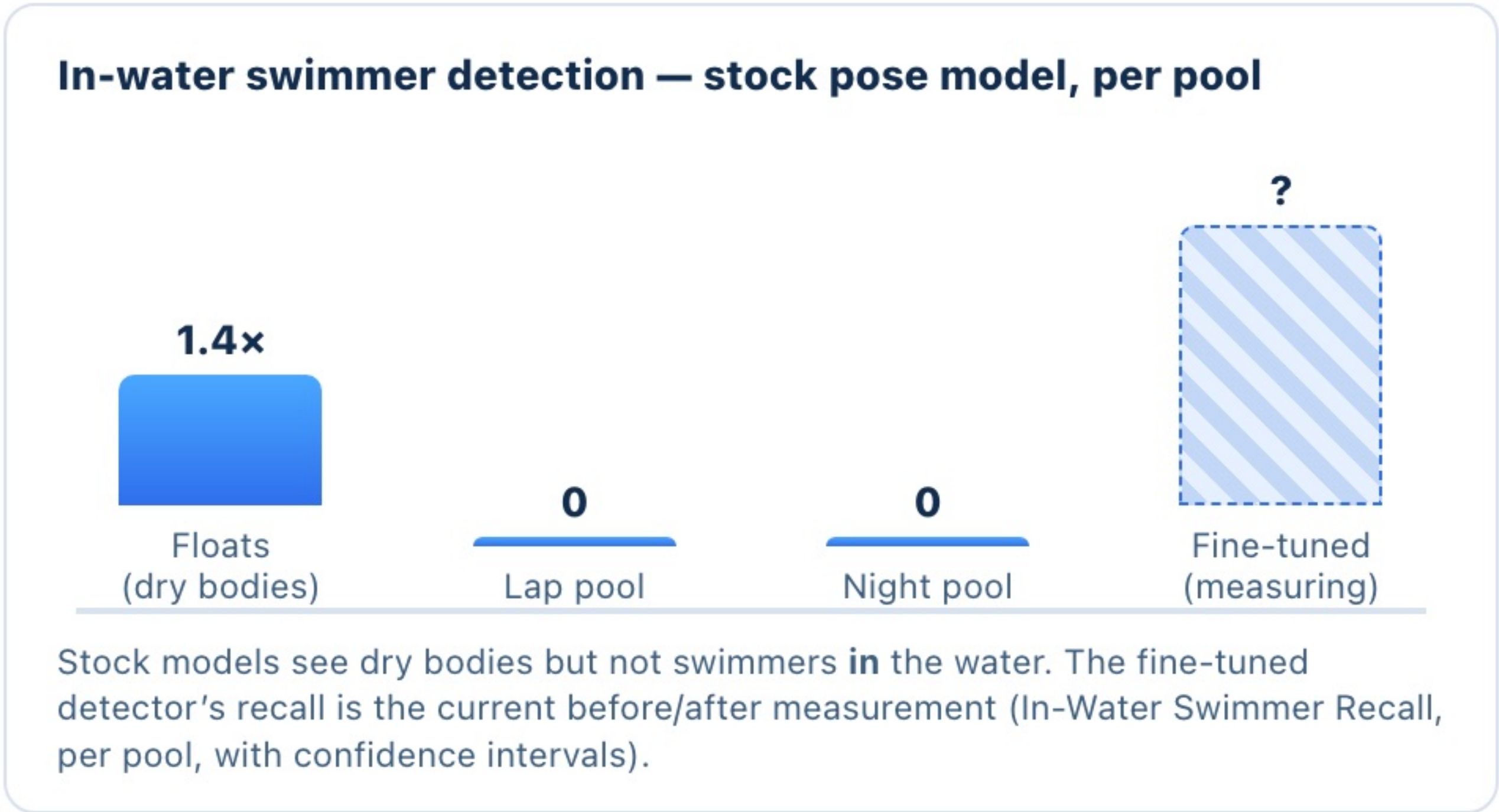


- 1

The pipeline works. 22 unit tests pass; verified end-to-end in a browser on demo and live video at ~9–26 fps. The engineering is not the bottleneck.
- 2

Detector-blindness (key finding). Across three real pools, stock MoveNet detected **0% of in-water swimmers** — a second model (COCO-SSD) agreed — while both found dry bodies on floats instantly. Night pool: 0 / 762 frames.
- 3

Real rescue footage. On fixed-camera pool CCTV of an actual child rescue, the submersion trigger fired — “check the water.”



Impact & Future Work

Now: fine-tune a YOLO swimmer detector on licensed pool imagery and publish In-Water Swimmer Recall (stock vs. fine-tuned, per pool). **Next:** hand-label the team’s own overhead pool footage to match a real mount, then measure across a YMCA, a hotel, and a neighborhood pool — the cross-pool number nobody else can produce.

Pharos detects stillness and submersion — not “drowning,” which no camera can truly know. It **assists** a lifeguard’s or parent’s supervision; it never replaces it.